

Computer Modeling of Factors Affecting Species Extinction

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BRIEF ABSTRACT

In order to explore mathematical relationships involved in species extinction, a computer simulation was developed that simulates a population of hypothetical animals with life history factors such as birth rate, life span, and dispersal rate. Mortality in the model is due to habitat fragmentation and random events. Experiments were performed to explore the effects of various life history factors on species extinction. The interrelationships among habitat fragmentation, dispersal rate, and random mortality events were also investigated.

Do I need more? Results?

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ABSTRACT

Florida and Ohio have recently adopted a ranking system to set conservation priorities for rare and endangered animal species, with points assigned for various factors that may increase the likelihood of extinction. However, there is currently no quantitative model underlying that system. In order to explore mathematical relationships involved in species extinction, a computer simulation was developed. This simulation models a population of a hypothetical animal species on a two-dimensional grid. The simulation can model life history factors such as life span, number of offspring per year, and dispersal rate, as well as habitat fragmentation and random, short-term mortality events. The simulation was used to compare some factors used in the Florida ranking system and also to explore other relationships between life history factors and species extinction. It was found that reproductive rate had a greater influence on species survival than did starting population size under the conditions of this model. Life span had practically no effect on survival. Increased dispersal had a negative effect on survival when a habitat was fragmented, and a non-linear positive effect when a habitat was subjected to random, short-term mortality events. When both random mortality events and habitat fragmentation were present, there appeared to be an optimal dispersal rate for best chance of survival. The interaction among these processes is currently being explored further.

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Abstract: Florida and Ohio have recently adopted a ranking system to set conservation priorities for rare and endangered animal species, with points assigned for various factors that may increase the likelihood of extinction. However, there is currently no quantitative model underlying that system. In order to explore mathematical relationships involved in species extinction, a computer simulation was developed. This simulation models a population of a hypothetical animal species on a two-dimensional grid. The simulation can model life history factors such as life span, number of offspring per year, and dispersal rate, as well as habitat fragmentation and random, short-term mortality events. The simulation was used to compare some factors used in the Florida ranking system and also to explore other relationships between life history factors and species extinction. It was found that reproductive rate had a greater influence on species survival than did starting population size under the conditions of this model. Life span had practically no effect on survival. Increased dispersal had a negative effect on survival when a habitat was fragmented, and a non-linear positive effect when a habitat was subjected to random, short-term mortality events. When both random mortality events and habitat fragmentation were present, there appeared to be an optimal dispersal rate for best chance of survival. The interaction among these processes is currently being explored further.

Problem: In order to set conservation priorities, agencies must know the relative influence of life history factors and sources of mortality on the probability of species extinction. A particularly interesting factor is the effect of dispersal behavior of a species. One can ask whether a higher dispersal rate increases or decreases the likelihood of extinction. In this example, how does the effect of mortality caused by fragmentation differ from that of mortality caused by short-term random events? A computer model offers a way to test the effects of these factors under various conditions, and lets one identify relationships among the factors.

Hypothesis: A computer model will show a positive relationship between dispersal rate and species survival in the face of short-term random mortality events.

Introduction: Conservation agencies have limited resources and must decide which species are most in need of protection. In 1990, Millsap et al. developed a point system to rank endangered animal species in Florida, and Ohio has recently adopted it. Various life history factors are considered in the ranking process, based on the influences they are thought to have on the likelihood of extinction. However, the system was not based on a quantitative model.

On the suggestion of Dr. Winfried Just at Ohio University, I have developed a computer model in order to explore the influences of some life history factors on extinction. I initially intended to use my model to compare the relative importance of factors in the Florida point system. But, as it turned out, many factors in the Florida system were difficult or impossible to simulate. Therefore, I modeled two factors from the Florida system and then went on to explore the effects of other life history factors on extinction.

The life history factors I modeled are reproductive rate, life span, starting population size, dispersal rate, and susceptibility to edge effect. Edge effect is the negative effect of disturbed habitat on adjacent undisturbed habitat. Causes of mortality in my model are random destructive events and habitat loss through fragmentation.

Procedure: I have constructed a computer program in a language called REALBasic. I decided to use REALBasic for a number of reasons. First, it is fully object oriented, which allowed me to construct classes and objects, letting me manage my simulation effectively. Second, REALBasic is excellent for constructing functional interfaces, and it let me create a new interface tool for drawing a grid on the screen. Third, I had previous experience with REALBasic, and I knew it would serve my needs effectively.

The computer program I constructed is a simulator that models a population of animals on a yearly basis using defined blocks of land called compartments. Each compartment is an object in the computer's memory and contains a number of variables. A compartmentalized model (Keen and Spain, 1992) is very useful for simulating a large population of individuals with limited computing power and time. The model uses a number of compartments, each with its own population of individuals. Unlike some models that have each individual as an object in the computer's memory, compartmentalized modeling uses algebraic equations to relate groups of individuals in a compartment. In my model, each compartment has two groups of individuals: those under reproductive age (group 1), and those of reproductive age (group 2). Life history factors of an animal are converted into functions that are used to relate the two groups. For example, when modeling a hypothetical animal that produces approximately four offspring every year and reaches reproductive at age five, I would use a function that would transfer $1/5$ of group 1 to group 2 yearly. That is because if the animals take five years to mature, every year $1/5$ of those animals must mature, and therefore must be moved to the reproductive group, group 2. I would also yearly create a number of new animals in group 1 equal to four times group 2, because the individuals of reproductive age in group 2 each have an average of four offspring yearly.

Another factor I can control in the model is death rate. A death rate is applied yearly to the mature individuals. It can either be applied yearly to the pre-reproductive age individuals or the same mathematical result can be achieved by applying the death rate all at once when an individual is born. For example, a possible life table might have 60% of the offspring die by the end of the first year, then 40% by the end of the second, then 25%, then 15%, then 10% for every year after that. Given this life table, the model would be set up so that a single death rate is applied to all offspring; in this case, only 13.8% of offspring would reach maturity (at age five). A 10% death rate would be applied yearly once the individuals reach reproductive age.

Each compartment has a maximum carrying capacity, which defines how many individuals the compartment can support. This capacity represents a combination of the amount of food available in a region and the size of the region. The simulation is capable of generating a randomly fluctuating carrying capacity, as would happen in real life. However, this feature can be disabled when not required.

Also included in the model is dispersal. Each compartment is assigned a position in a grid. This grid holds the entire population, and each square is a sub-population. Currently, the grid is 10 by 10, allowing for 100 square compartments, but this number can easily be changed. The model can be configured to cause a certain percentage of each square's population to migrate to adjacent squares each year.

There are three ways in which the simulation can be monitored as it runs. First, the grid itself is shown on the screen, and each square is colored a shade of gray, where black is completely empty, and white is at maximum carrying capacity. Second, the user can select a single square and see the data for that square. Third, the simulation provides a graph of the total population of the entire grid over time. This graph helps identify trends in population size.

An important aspect of the simulator is the ability to create barriers. These barriers are areas that are so foreign to the simulated animals that they cannot live there. There are two types of barrier. The first type is a solid barrier that cannot be penetrated and can therefore be used to isolate an area of the grid or to experiment with habitat loss by turning inhabitable squares into solid, uninhabitable areas. The second type of barrier is passable but not inhabitable. Animals can migrate through but not live in it, and a user-defined percentage of them die if they attempt to cross it. (This might correspond to a parking lot or road in the real world.) Both types of barriers increase the death rate of the surrounding squares: This increase, which can be disabled, represents the negative edge effect seen at the boundaries of disturbed habitats. This feature of the model applies best to animals that do not survive well in edge habitat. (Wilcove et al., 1986).

The barriers were used to simulate fragmentation of the habitat. To fragment the grid, a certain percentage of randomly selected squares can be turned into barriers. Fragmentation hurts the animal population because the new barriers not only kill animals in them when they were created, but also decreased the survival rates of surrounding grid squares.

Random events occur in the model and represent both natural and unnatural mortality events. In nature, such events could include fires or floods or human-caused random mortality such as hunting or lumbering.

To test the relationship between a life history factor and extinction, the factor is varied experimentally. For each value set for the factor, the degree of habitat fragmentation is increased progressively until extinction occurs. This is done separately for each life history factor.

Results: (See graph 1) A positive relationship between reproductive rate and survival was found. In other words, the higher the reproductive rate, the more likely it is that the species will survive fragmentation. This is not surprising, but getting an expected result increases confidence in the simulation. Reproductive rate was tested at three different dispersal rates. It was found that as dispersal rate increased, likelihood of survival

decreased. This is probably occurring because as dispersal increases, more animals wander into the fragmentation barriers and die. (See graph 4)

A positive relationship was also found between starting population size and species survival (see graph 2), but the strength of the relationship does not seem to be as great as between reproductive rate and survival. As before, when fragmentation was involved, dispersal had a negative effect.

Life span was also tested to see how it affects species survival. Interestingly, there appeared to be almost no change in survival rate with increased life span. (see graph 3)

The effects of dispersal rate on animals' survival of random short-term mortality events was also explored. This was tested in the same way that fragmentation was, increasing the probability of random events until extinction occurred. The resulting curve starts out extremely steeply, sloping downward, and then levels off. (See graph 5)

When dispersal rate was tested with both random events and fragmentation applied as mortality factors, another interesting curve was found. (See graph 6)

Discussion:

The curve that relates habitat fragmentation, random mortality events, and dispersal rate appears to show an optimal dispersal rate for best chance of survival. This suggests an intriguing interrelationship between fragmentation, dispersal, and random mortality events. I plan to explore this relationship further. The curve may change shape when different fragmentation levels are applied.

My results suggest that reproductive rate has a greater influence on survival than does starting population size. This is in contrast with the Florida system, which gives more weight to population size. However, population size may be a very important indicator of other problems even if it does not have a strong direct effect on survival itself.

Conclusion: Simulations are simplifications of the real world and therefore are never completely representative of it. This simulation attempts to simplify an extremely complicated process--extinction--and undoubtedly loses some accuracy. However, it is looking for trends and relationships, not attempting to predict a specific outcome, so it should still be informative.

Under the conditions of this model, my results suggest that reproductive rate has a greater influence on survival than starting population size does. More generally, my experiment suggests that a simulation of this sort can help inform and improve ranking systems such as the one used in Florida and Ohio.

My results also suggest that the Florida ranking system should include susceptibility to edge effect, current level of fragmentation, and dispersal rate. All three factors appear to be important in species survival, and are not included in the Florida ranking system.

Finally, there appears to be an interrelationship among dispersal, fragmentation, and random mortality events. My experiments support my hypothesis that a positive relationship would be found between dispersal rate and species survival in the face of short-term random mortality events. Though a direct linear relationship was not found, dispersal rate did initially have a positive effect on species survival. On the other hand, when mortality was due to fragmentation alone, increased dispersal rate had a negative effect on survival. When both fragmentation and random, short-term mortality events were introduced, they interacted to produce a curved graph. A point was found that probably represents the best possible dispersal rate for the particular fragmentation level and random event probability tested.

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